

Chapter 16: Thermodynamics

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CHM111
Lecture 16



Thermodynamics: Using the power of *Heat* to do *Work*

θέρμη [t^hér.mɛ:]

heat

noun – Ancient Greek

δύναμις [dý.na.mis]

power, might, strength

noun – Ancient Greek

Thermodynamics describes the relationship of *heat*, a flow of energy, to *work*, the ability to move things.

Thermodynamics studies the limits on how much thermal energy can be transformed into organized mechanical or electrical energy that we can *actually use to perform tasks*.

Thermodynamics

Thermodynamics extends our study of energy (Ch. 5) to answer a new question:
Which processes actually happen on their own?

- ▶ Chapter 5 asked: *how much energy* is exchanged?
- ▶ Chapter 16 asks: *which direction* does a process go?
 - ▶ The answer requires two quantities — **entropy** (S) and **free energy** (G).

Chapter Overview

Topics Covered in Chapter 16

1. **Spontaneity** — direction of change; dispersal of matter and energy
2. **Entropy** (S) — microstates; Boltzmann's equation $S = k \ln W$
3. **The Second and Third Laws** — $\Delta S_{\text{univ}} \geq 0$; standard entropies
4. **Free Energy** (G) — $\Delta G = \Delta H - T\Delta S$; equilibrium and temperature dependence

Part I

Spontaneity

Outline of Part I: Spontaneity

Spontaneous and Nonspontaneous Processes

Outline of Part I: Spontaneity

Spontaneous and Nonspontaneous Processes

What Is a Spontaneous Process?

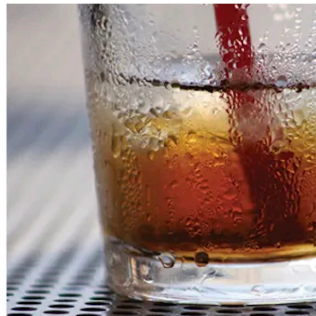
Definition: Spontaneous Process

- ▶ A **spontaneous process** occurs *without any continuous outside intervention*.
 - ▶ A **nonspontaneous process** requires a *sustained input of energy* from the surroundings to take place.
-
- ▶ A rock rolls downhill *spontaneously*; it does not roll uphill by itself.
 - ▶ Hot coffee cools in a room; it never warms itself back up.
 - ▶ Iron rusts in moist air; rust does not un-rust into iron and oxygen.

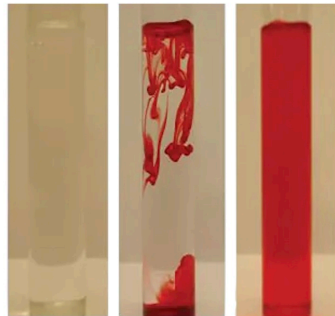
Everyday Spontaneous Processes



(a)



(b)



(c)

(a) An ice cube melts in a warm room. (b) A hot beverage cools toward room temperature. (c) A drop of dye disperses through water. Each occurs **on its own**; the reverse would demand outside intervention.

Spontaneous $\neq$ Fast

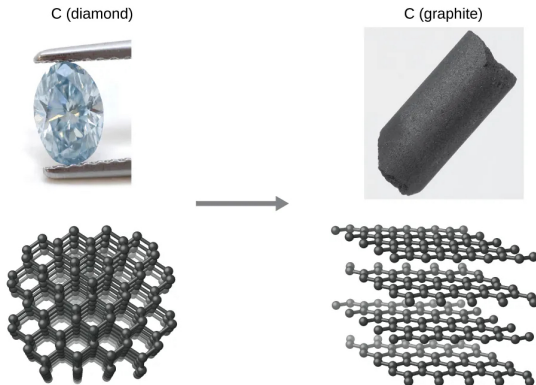
Common Misconception

“Spontaneous” does **not** mean fast. It only means that a process *tends* to proceed in a certain direction *on its own*.

Thermodynamics vs. Kinetics

- ▶ **Thermodynamics** tells us *whether* a process is favorable.
- ▶ **Kinetics** (Ch. 12) tells us *how fast* it happens.
- ▶ A process may be spontaneous yet immeasurably slow at room temperature.

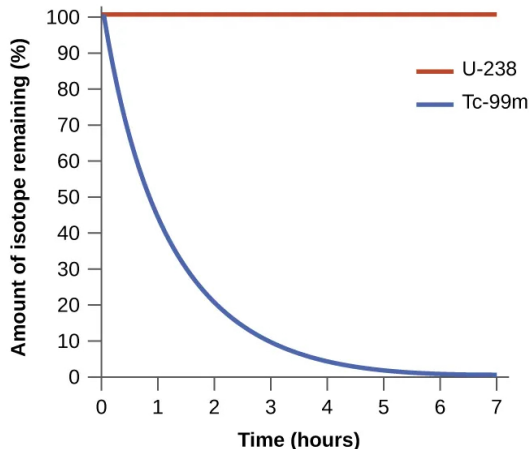
Example of Rate v. Spontaneity: Diamond $\rightarrow$ Graphite



The conversion of **diamond** $\rightarrow$ **graphite** at 1 atm and 25 °C is **spontaneous** — but takes billions of years.

- ▶ Graphite is the more stable allotrope of carbon at STP.
- ▶ Diamond is only *kinetically* trapped by a huge activation barrier.
- ▶ “Diamonds are forever” is kinetics, not thermodynamics.

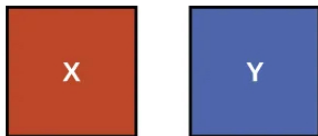
Example of Rate v. Spontaneity: Isotope Decay



Both nuclear decays are **spontaneous**.
Only *how fast* they proceed differs.

- ▶ Tc-99m: half-life ≈ 6 h (fast).
- ▶ U-238: half-life $\approx 4.5 \times 10^9$ yr (slow).
- ▶ Same word, wildly different timescales.

Heat Flows Spontaneously from Hot to Cold



$$T_X > T_Y$$

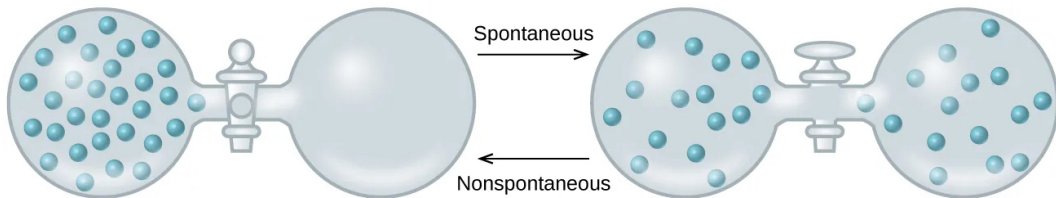


X and Y in contact

When two objects at different temperatures are placed in thermal contact, heat flows **spontaneously** from the hot object (X) to the cold object (Y) until $T_X = T_Y$.

The reverse — heat flowing from cold to hot *on its own* — is **never observed**.

Dispersal of Matter: Free Expansion



Opening the stopcock lets an ideal gas spontaneously spread throughout both flasks. No energy is exchanged with the surroundings ($\Delta U = 0$), yet the process proceeds $\because$ **matter becomes more *dispersed*.**

Dispersal Drives Spontaneous Change

Two Kinds of Dispersal

- ▶ **Matter** spreads out (gases fill their containers, solutes mix).
 - ▶ **Energy** spreads out (hot bodies cool, kinetic energy is shared).
-
- ▶ Neither dispersal type violates the First Law: energy is still conserved.
 - ▶ But the First Law does not explain which direction a process will occur spontaneously.

We need a new state function — **entropy** (S) — that measures dispersal.

Check Your Understanding

A cup of hot cocoa sits on a table and cools to room temperature.

- (a) Is the cooling process spontaneous?
- (b) Would the reverse process ever occur on its own?

Come to lecture to see the answer.

Check Your Understanding

Diamond is thermodynamically unstable relative to graphite at room temperature. Why do we still see diamonds in nature and in jewelry stores?

Come to lecture to see the answer.

Part II

Entropy

Outline of Part II: Entropy

Entropy and Microstates

Predicting the Sign of Entropy Change

Outline of Part II: Entropy

Entropy and Microstates

Predicting the Sign of Entropy Change

Rudolf Clausius coined the term *entropy* to parallel *energy*:

- ▶ *energy* tells us about the ability to cause change.
- ▶ *entropy* tells us about the *natural tendency* of *that energy* to become more widely dispersed among the many possible microscopic arrangements of the system.

Energy and Entropy

ἐνέργεια [e.nér.ge:̌.a]

noun – Ancient Greek

work within

$\text{ev}(\text{en}) = \text{“within”}$

ἔργον (ergon) = “work,” “deed,” or “action”

ἐντροπία [en.tro.pí.a:]

noun – Ancient Greek

internal transformation

ẽv (en) = “within”

τροπή (tropē) = “turning,” “change,” “transformation”



A

Ex

A

Entropy in Thermodynamics

For a reversible process, thermodynamics defines the change in entropy by

$$\Delta S = \int \frac{\delta Q_{\text{rev}}}{T}$$

where

- ▶ ΔS = change in entropy,
- ▶ δQ_{rev} = heat transferred reversibly,
- ▶ T = absolute temperature.

Entropy has units of J/K.

Experimentally, entropy is a property of a system that

- ▶ changes when heat is added or removed,
- ▶ helps determine the direction of natural processes,
- ▶ never decreases for an isolated system.

A Fundamental Question

Thermodynamics introduces this quantity called *entropy*.

But thermodynamics does not tell us *what entropy actually is*.

What does entropy mean at the *microscopic* level?

Defining Terms: Microstates v. Macrostates

Microstate

A specific arrangement of every particle
— positions, momenta, orientations.

A microscopic “snapshot” of the system.

Multiplicity W refers to the *number of microstates* corresponding to a *particular macrostate* of a thermodynamic system.

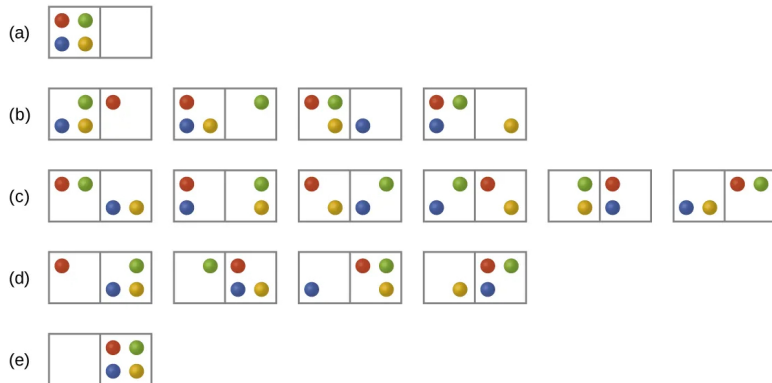
A macrostate with more microstates (W) is **overwhelmingly more likely** to be observed.

Macrostate

The state characterized by macroscopic variables (P , V , T , n).

Many microstates share the same macrostate.

Counting Microstates: Particles in Two Boxes



Four particles are distributed in a two-box system. Each possible microstate is shown above. The **even split (c)** has the greatest number of microstates (6 of 16 total), $\therefore$ it is the *most probable macrostate*.

Extremes (a, e) have only 1 microstate each.

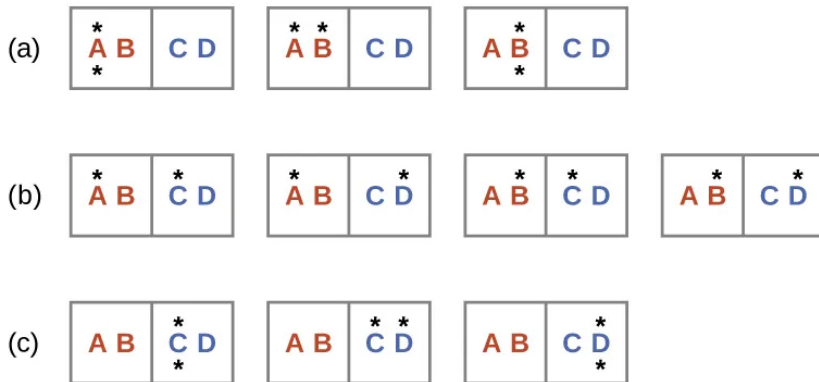
Probability of Each Distribution

Four Distinguishable Particles in Two Boxes

$$P(\text{even split}) = \frac{6}{16} = 37.5\%, \quad P(\text{all one side}) = \frac{1}{16} = 6.25\%$$

- ▶ With 4 particles, the “all on one side” outcome is rare but not impossible.
- ▶ With $N = 6.022 \times 10^{23}$ particles (1 mole), $P(\text{all on one side}) \approx 2^{-N}$ — effectively zero.
- ▶ That is why gases **never** spontaneously compress themselves into one flask.

Distributing Energy Quanta Among Particles



Two energy quanta (*) are distributed among four particles (A, B, C, D) in two boxes. Configuration (b), with one quantum on each side, has the most microstates — energy is most likely to be **dispersed** across both boxes.

Quantum Mechanics Describes States

Quantum mechanics tells us that every isolated system has discrete energy levels (eigenstates)

$$E_0, E_1, E_2, \dots$$

Each energy eigenstate is a *possible quantum state* of the system.

At absolute zero, the system occupies its quantum ground state. If the ground state is *non-degenerate*, $W = 1$, so $S = 0$, consistent with the Third Law of Thermodynamics.

A degenerate ground state can produce residual entropy, a quantum effect beyond the scope of this discussion.

Degeneracy in the Ground State

If the ground state is *non-degenerate*, then there is only one quantum microstate available at $T = 0$, so

$$W = 1,$$

$$S = k_B \ln W = 0.$$

If the ground state is *degenerate*, then multiple distinct quantum microstates have the same minimum energy, giving

$$W > 1,$$

and therefore a residual entropy,

$$S = k_B \ln W > 0.$$

Why Do We Need Multiplicity W ?

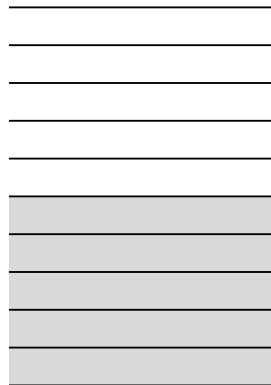
Consider a system at very low temperature. Only the lowest-energy states are likely to be occupied.

As we add thermal energy, the system can explore more of its available quantum states.

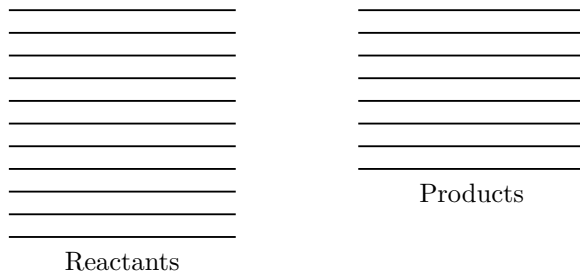
- ▶ More states become energetically accessible.
- ▶ More microscopic arrangements become possible.
- ▶ The number of accessible microstates W increases.

Quantized Energy Level Diagram of a Molecular System

- ▶ Each horizontal line is an *allowed* molecular energy level.
- ▶ The lowest level is the *ground state*; higher levels are *excited states*.
- ▶ At any temperature, molecules can occupy these levels according to the Boltzmann distribution (the shaded region illustrates *occupied states*).
- ▶ The distribution of molecules across excited states determines the system's *thermodynamic properties*.
- ▶ The number and spacing of accessible energy levels determine the number of *microstates* (W).
- ▶ More numerous accessible microstates correspond to greater entropy.



Enthalpy Change and Entropy Change



The ground state corresponds to the value of the *enthalpy* H
 $\Delta H > 0$ means products' ground state is higher ($T = 0\text{K}$)
 ΔH reflects changes in the strength of the chemical bonds.

Position of the ground state tells you ΔH
 Number of accessible energy levels/microstates tells you ΔS

An Important Clarification

Increasing the temperature does not *create* new quantum energy levels.

The discrete, allowed energy levels are fixed by the system's quantum mechanics.

Heating the system simply allows it to occupy a larger fraction of those already-existing states.

Where Does Temperature Enter?

Temperature is not itself an energy. Instead, $k_B T$ defines the *characteristic thermal energy scale*.

The Boltzmann factor $P(E) \propto e^{-E/(k_B T)}$ determines the *thermal population of energy levels*.

Let ΔE denote the energy gap between two quantum states.

If $\Delta E \lesssim k_B T$, the excited state becomes thermally accessible.

If $\Delta E \gg k_B T$, the excited state is rarely occupied.

Temperature Determines Accessible States

As temperature increases, $k_B T$ increases.

Therefore,

- ▶ more excited states become populated,
- ▶ more quantum states become thermally accessible,
- ▶ the number of accessible microstates increases.

Higher temperature $\therefore$ corresponds to a greater multiplicity (a larger value of W).

The greater the multiplicity of a macrostate is, the greater its entropy is.

Why Should Entropy Count Microstates?

- ▶ Question
 - ▶ Why should a macroscopic state depend on microscopic arrangements?
- ▶ Evidence
 - ▶ Many distinct particle arrangements produce identical pressure, volume, and temperature.
 - ▶ Equilibrium consistently corresponds to the most frequently observed macrostate.
- ▶ Reasoning
 - ▶ $\therefore$ A macrostate is realized whenever the system occupies any compatible microscopic arrangement.
 - ▶ $\therefore$ More compatible *microstates* make that macrostate more likely to appear.
- ▶ Takeaway
 - ▶ Probability depends on counting microscopic possibilities.

Counting Combined Possibilities

Suppose two independent systems can each occupy several microscopic arrangements.

System 1	System 2
A	X
B	Y
C	Z
D	

A, B, C, D and X, Y, Z each represent one microscopic arrangement.

System 1 has $W_1 = 4$ arrangements and System 2 has $W_2 = 3$ arrangements.

Every arrangement of System 1 can occur with every arrangement of System 2:
 (A, X); (A, Y); (A, Z); (B, X); ...

$$\therefore W_{\text{total}} = 4 \times 3 = 12.$$

Why Do the Numbers Multiply?

To specify the microscopic state of the combined system, we must specify

- ▶ which arrangement System 1 occupies, and
- ▶ which arrangement System 2 occupies.

If System 1 has W_1 possibilities and System 2 has W_2 possibilities, then each of the W_1 arrangements can be paired with each of the W_2 arrangements.

$$W_{\text{total}} = W_1 W_2.$$

The number of microscopic arrangements *multiplies* for independent systems.

The Need for a Logarithm

Experiment shows that entropy is an extensive property. For two independent systems,

$$S_{\text{total}} = S_1 + S_2.$$

But microscopic counting gives

$$W_{\text{total}} = W_1 W_2.$$

We therefore need a function that converts products into sums:

$$S = k \ln W,$$

because

$$\ln(W_1 W_2) = \ln W_1 + \ln W_2.$$

Why Must Entropy Be A Logarithm?

- ▶ Question
 - ▶ Why is entropy proportional to $\ln W$ rather than W ?
- ▶ Evidence
 - ▶ Independent systems satisfy $W_{\text{tot}} = W_1 W_2$.
 - ▶ Measured entropy doubles when identical independent systems are combined.
- ▶ Reasoning
 - ▶ $\therefore$ Independent possibilities multiply while extensive properties must add physically.
 - ▶ $\therefore$ Only the logarithm converts multiplied possibilities into additive *entropy*.
- ▶ Takeaway
 - ▶ Additivity uniquely determines the logarithm.

Why Boltzmann's Constant?

Temperature first enters microscopic physics through the thermal energy scale $k_B T$.

This energy scale determines which quantum states become thermally accessible and therefore how many microstates contribute to the system.

Entropy measures this multiplicity through $\ln W$.

Since entropy has units of J/K while $\ln W$ is dimensionless, $S = k_B \ln W$.

The same constant that relates temperature to microscopic energy also connects the multiplicity of quantum states to thermodynamic entropy.

The Boltzmann Bridge: T , W , S

$$T \longrightarrow k_B T \longrightarrow \text{accessible quantum states} \longrightarrow W \longrightarrow S = k_B \ln W$$

- ▶ Temperature sets the microscopic thermal energy scale.
- ▶ The thermal energy scale determines which quantum states are appreciably populated.
- ▶ Accessible quantum states determine the multiplicity W .
- ▶ Entropy is the logarithmic measure of that multiplicity.
- ▶ Boltzmann's constant bridges macroscopic temperature and microscopic quantum physics.

Entropy (S) – SI Unit: J/K

Definition

Entropy is a state function measuring the number of microscopic arrangements (W) available to a system:

$$S = k_B \ln W$$

where $k_B = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$ is Boltzmann's constant.

- ▶ More accessible arrangements $\Rightarrow$ higher entropy.
- ▶ Systems tend toward states with the most microstates.
- ▶ Entropy is a *measure of dispersal*.

Entropy Change ΔS

Change in Entropy

$$\Delta S = S_f - S_i = k_B \ln \frac{W_f}{W_i}$$

- ▶ $\Delta S > 0$: system has more microstates in the final state (more dispersed).
- ▶ $\Delta S < 0$: system has fewer microstates in the final state (more concentrated).
- ▶ $\Delta S = 0$: same number of microstates.

Entropy is a **state function**: ΔS depends only on initial and final states.

Why Does Nature Have a Preferred Direction?

- ▶ Question
 - ▶ Why do reversible molecular laws produce irreversible everyday change?
- ▶ Evidence
 - ▶ Molecular collisions obey time-reversible mechanical laws.
 - ▶ Melting and mixing occur naturally; exact reverses are never observed.
- ▶ Reasoning
 - ▶ Countless molecules overwhelmingly evolve toward *more numerous microscopic arrangements*.
 - ▶ $\therefore$ Macroscopic direction emerges statistically despite reversible individual collisions.
- ▶ Takeaway
 - ▶ Statistics create macroscopic direction from reversible dynamics.

Why Do Different Processes Share One Direction?

- ▶ Question
 - ▶ Why do melting, cooling, and mixing all proceed naturally?
- ▶ Evidence
 - ▶ Ice melts, hot coffee cools, and dye disperses without guidance.
 - ▶ Their exact reverses require deliberate external intervention.
- ▶ Reasoning
 - ▶ $\therefore$ Each process spreads energy or particles across vastly more accessible molecular arrangements.
 - ▶ $\therefore$ We can use one statistical mechanism to explain these apparently different phenomena.
- ▶ Takeaway
 - ▶ *Accessible molecular arrangements* determine natural direction.

Why is it that Energy Alone Cannot Predict Direction?

- ▶ Question
 - ▶ Why is the First Law unable to identify spontaneous change?
- ▶ Evidence
 - ▶ A spontaneous process and its reverse can conserve total energy equally.
 - ▶ Energy conservation is unchanged when time is reversed.
- ▶ Reasoning
 - ▶ $\therefore$ Conserved energy tracks quantity transferred, not molecular arrangement probability.
 - ▶ $\therefore$ Another state function is required to distinguish direction.
- ▶ Takeaway
 - ▶ Conservation alone cannot determine process direction.

Why Does Free Expansion Emphasize $\Delta U = 0$?

- ▶ Question
 - ▶ Why does gas expand spontaneously even when internal energy remains unchanged?
- ▶ Evidence
 - ▶ During ideal-gas free expansion, $\Delta U = 0$.
 - ▶ The gas still fills the available volume spontaneously.
- ▶ Reasoning
 - ▶ $\therefore$ Larger volume permits enormously more molecular position arrangements without changing molecular energy.
 - ▶ $\therefore$ Direction is driven by arrangement availability rather than energy decrease.
- ▶ Takeaway
 - ▶ Arrangement availability can drive change without energy change.

Why Can A Spontaneous Change Fail To Occur?

- ▶ Question
 - ▶ Why does diamond persist although graphite is thermodynamically favored?
- ▶ Evidence
 - ▶ Diamonds remain unchanged for geological times under ordinary conditions.
 - ▶ Heating greatly accelerates conversion toward graphite.
- ▶ Reasoning
 - ▶ $\therefore$ Strong bonding requires rare high-energy rearrangements before lower-energy structures become accessible.
 - ▶ $\therefore$ *Thermodynamic tendency* and *kinetic rate* are independent properties.
- ▶ Takeaway
 - ▶ *Driving force* and *rate* answer different questions.

Why Assume Equal Microstate Probability?

- ▶ Question
 - ▶ Why are all accessible microstates treated equally?
- ▶ Evidence
 - ▶ Equilibrium experiments reveal no preferred arrangement among equivalent microstates.
 - ▶ Only conserved macroscopic constraints affect observed equilibrium populations.
- ▶ Reasoning
 - ▶ $\therefore$ Persistent microscopic preferences would produce repeatable macroscopic asymmetries absent experimentally.
 - ▶ $\therefore$ Equal a priori probability makes counting predict equilibrium observations.
- ▶ Takeaway
 - ▶ Counting requires unbiased microscopic sampling.

Why Does Large N Create Irreversibility?

- ▶ Question
 - ▶ Why does a statistical preference become macroscopic certainty?
- ▶ Evidence
 - ▶ Four particles occupy one side with probability $1/16$.
 - ▶ For N particles, complete localization occurs with probability 2^{-N} .
- ▶ Reasoning
 - ▶ $\therefore$ Each added particle doubles possible arrangements while only one remains completely localized.
 - ▶ $\therefore$ For $N \approx 10^{23}$, reverse macrostates become effectively impossible to observe.
- ▶ Takeaway
 - ▶ Large numbers create *empirical certainty*.

Why Does Boltzmann's Constant Have Units?

- ▶ Question
 - ▶ Why must k_B have units of J/K?
- ▶ Evidence
 - ▶ $\ln W$ has no physical units.
 - ▶ Thermodynamic entropy is experimentally measured in J/K.
- ▶ Reasoning
 - ▶ $\therefore$ Microscopic counting must reproduce the established thermodynamic entropy scale.
 - ▶ $\therefore k_B$ converts dimensionless counting into measurable entropy.
- ▶ Takeaway
 - ▶ Boltzmann's constant bridges microscopic and macroscopic descriptions.

Why Do Matter And Energy Disperse Similarly?

- ▶ Question
 - ▶ Why are matter dispersal and energy dispersal one phenomenon?
- ▶ Evidence
 - ▶ Mixing particles and sharing energy both increase accessible microstates.
 - ▶ The same two-box counting predicts both processes.
- ▶ Reasoning
 - ▶ $\therefore$ Both changes enlarge the number of microscopic arrangements satisfying the same macroscopic constraints.
 - ▶ $\therefore$ Entropy measures accessible possibilities regardless of whether matter or energy spreads.
- ▶ Takeaway
 - ▶ Microstate counting unifies all spontaneous dispersal.

Entropy and Microstates

Predicting the Sign of Entropy Change

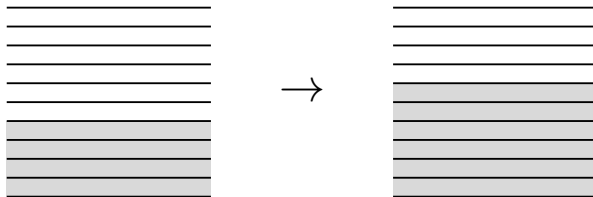
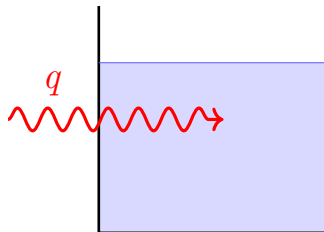
Entropy of a substance **increases** when:

- Temperature rises (more thermal microstates accessible).
- Volume increases (more positional microstates).
- Phase changes: solid $\rightarrow$ liquid $\rightarrow$ gas.
- Moles of gas particles increase in a reaction.
- A pure solid or liquid dissolves in a solvent.
- A complex molecule replaces simpler ones (more vibrational modes).

Entropy S Relates to Temperature

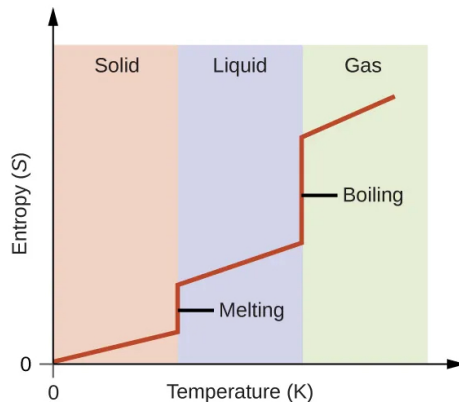
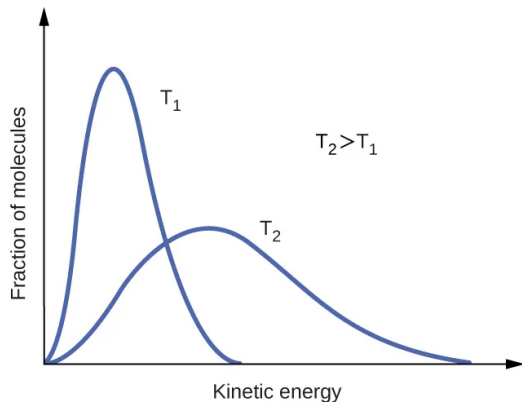
Entropy S measures the degree to which energy is dispersed in a system.

- ▶ Add a small amount of heat q to a system at temperature T .
- ▶ The entropy change ΔS is given by $\Delta S = \frac{q}{T}$ with units J/K.
- ▶ ΔS measures the *spreading out of heat* in the system resulting from the addition of q .



When q causes an increase in T , the heat spreads into higher energy levels (and the molecules occupy a wider range of thermally excited states).

Entropy S Relates to Temperature



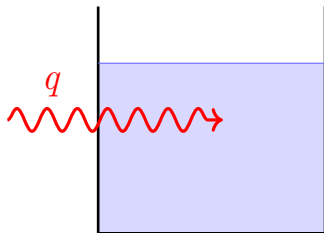
Left: At higher T , molecular kinetic energies are spread over a wider range of values — more thermal microstates are accessible.

Right: S rises with T , with sharp jumps at melting and boiling.

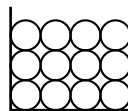
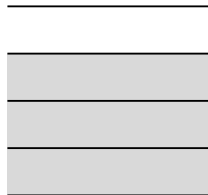
Why Does Temperature Increase Entropy?

- ▶ Question
 - ▶ Why does heating increase the number of accessible microstates?
- ▶ Evidence
 - ▶ Higher temperatures populate higher molecular energy levels.
 - ▶ Measured entropy increases as temperature rises.
- ▶ Reasoning
 - ▶ $\therefore$ Additional thermal energy allows molecules to occupy previously inaccessible translational, rotational, and vibrational states.
 - ▶ $\therefore$ More accessible energy levels multiply the total number of available microstates.
- ▶ Takeaway
 - ▶ Energy dispersal increases accessible microstates.

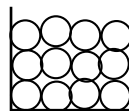
Phase Changes and Entropy



When q causes a *phase change* (melting, vaporization), then T remains constant, and the heat disrupts intermolecular forces, making molecules less constrained, which allows *more low-energy levels* to “open up”.

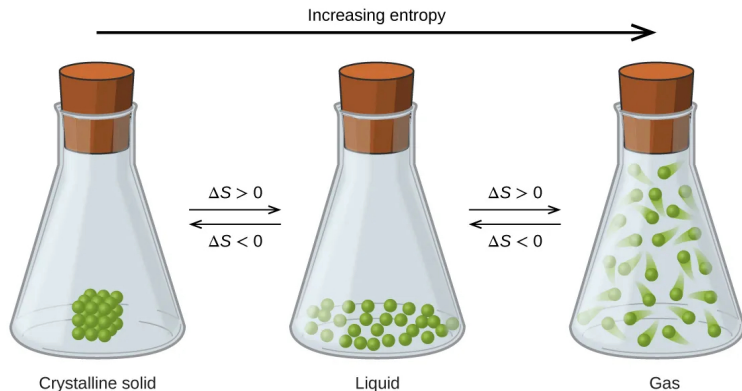


Molecules
in a Solid



Molecules
in a Liquid

Phase Changes and Entropy



Entropy increases dramatically going solid $\rightarrow$ liquid $\rightarrow$ gas: particles gain translational and rotational freedom, occupying vastly more microstates.

Reverse processes (condensation, freezing) have $\Delta S < 0$.

Predicting ΔS for Physical Changes

- ▶ Melting, vaporization, sublimation: $\Delta S > 0$.
- ▶ Freezing, condensation, deposition: $\Delta S < 0$.
- ▶ Heating a substance: $\Delta S > 0$.
- ▶ Compressing a gas: $\Delta S < 0$.

For a chemical reaction, look first at the **change in moles of gas**:

$$\Delta n_{\text{gas}} > 0 \Rightarrow \Delta S > 0; \quad \Delta n_{\text{gas}} < 0 \Rightarrow \Delta S < 0.$$

Example: Predicting the Sign of ΔS

Sign of ΔS

Predict the sign of ΔS for each process:

- (a) One mole liquid water boils to steam at 100 °C.
- (b) $\text{CO}_2(\text{s}) \longrightarrow \text{CO}_2(\text{g})$ (dry ice sublimates).
- (c) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2 \text{HCl}(\text{g})$.
- (d) $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2 \text{SO}_3(\text{g})$.

- (a) Liquid $\rightarrow$ gas: $\Delta S > 0$.
- (b) Solid $\rightarrow$ gas: $\Delta S > 0$.
- (c) $\Delta n_{\text{gas}} = 2 - 2 = 0$: $\Delta S \approx 0$ (small).
- (d) $\Delta n_{\text{gas}} = 2 - 3 = -1$: $\Delta S < 0$.

- Which change involves the largest positive ΔS ?
- Why is (c) close to zero even though a new substance forms?

Why Does Vaporization Increase Entropy So Much?

- ▶ Question
 - ▶ Why is the entropy increase much larger for boiling than melting?
- ▶ Evidence
 - ▶ Experimental ΔS_{vap} greatly exceeds ΔS_{fus} .
 - ▶ Gas molecules occupy far larger volumes than liquid molecules.
- ▶ Reasoning
 - ▶ $\therefore$ Vaporization releases molecules from neighboring constraints, creating vastly more translational arrangements than melting.
 - ▶ $\therefore$ The accessible microstate increase is much larger during vaporization.
- ▶ Takeaway
 - ▶ Translational freedom dominates entropy changes.

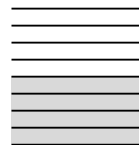
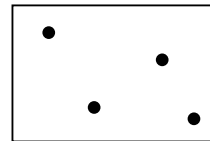
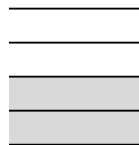
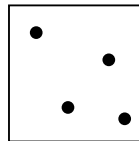
Entropy Increases with Volumetric Expansion

$\therefore$ Increasing the volume available to a system gives particles access to more possible positions, even if the number of particles remains unchanged.

Consider this:

- ▶ Number of particles N stays constant
- ▶ Volume V of the system increases
- ▶ More accessible microstates become available
- ▶ $\therefore$ Entropy rises with volume

$$S \propto \ln(V)$$



Why Does Expansion Increase Entropy?

- ▶ Question
 - ▶ Why does increasing volume raise entropy even without heating?
- ▶ Evidence
 - ▶ Doubling an ideal gas volume doubles each particle's accessible positions.
 - ▶ Measured ideal-gas entropy increases with volume.
- ▶ Reasoning
 - ▶ $\therefore$ Independent particles each gain additional positional choices, so total arrangements multiply across all particles.
 - ▶ $\therefore$ Entropy increases because positional microstates grow multiplicatively with volume.
- ▶ Takeaway
 - ▶ Spatial freedom multiplies microscopic possibilities.

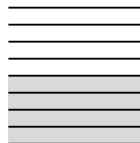
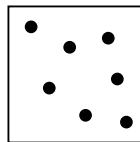
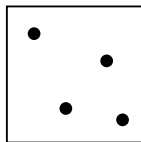
Entropy $\propto$ Number of Particles

$\therefore$ Entropy is related to substances' composition and internal structure, it is possible for a system's entropy to change without an addition of heat.

Consider this:

- ▶ Number of particles N in the system rises
- ▶ More particles mean more energy levels
- ▶ $\therefore$ Entropy rises with the number of particles

$$S \propto N$$



Why does Changing Gas Moles Strongly Affect ΔS_{rxn} ?

- ▶ Question
 - ▶ Why does changing gas moles strongly affect reaction entropy?
- ▶ Evidence
 - ▶ Reactions producing more gas usually have positive ΔS .
 - ▶ Changing solid or liquid moles often changes entropy much less.
- ▶ Reasoning
 - ▶ $\therefore$ Each gas molecule contributes many independent translational arrangements unavailable in condensed phases.
 - ▶ $\therefore$ Gas stoichiometry largely determines the reaction microstate count.
- ▶ Takeaway
 - ▶ Gas particles dominate accessible arrangements.

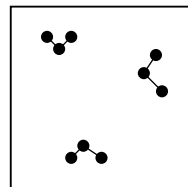
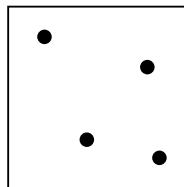
Entropy S Relates to Molecular Size

$\therefore$ Larger molecules possess more ways to store energy through rotational and vibrational motion, resulting in more accessible microstates.

Consider this:

- ▶ Number of particles N stays constant
- ▶ Molecules become larger and more complex
- ▶ More rotational & vibrational energy levels exist
- ▶ $\therefore$ Entropy rises with molecular size

$$S \propto \text{molecular complexity}$$

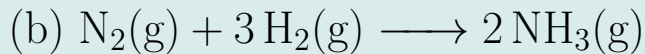


Why Do Larger Molecules Have Higher Entropy?

- ▶ Question
 - ▶ Why do heavier or more complex molecules have larger standard entropies?
- ▶ Evidence
 - ▶ Standard entropy generally increases with molecular mass and complexity.
 - ▶ Larger molecules possess more rotational and vibrational motions.
- ▶ Reasoning
 - ▶ $\therefore$ *Additional* atoms create *more* independent molecular motions with closely spaced accessible energy levels.
 - ▶ $\therefore$ More molecular motions multiply the number of accessible microstates.
- ▶ Takeaway
 - ▶ Additional motions multiply microscopic possibilities.

Check Your Understanding

Predict the sign of ΔS for each process:



(c) Rubbing alcohol evaporating from your skin

Come to lecture to see the answer.

Part III

The Second and Third Laws of Thermodynamics

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

The System, the Surroundings, and the Universe

For any process we identify three regions:

- ▶ **System** — the reaction or region of interest.
- ▶ **Surroundings** — everything else that can exchange heat with the system.
- ▶ **Universe** = system + surroundings.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

The Second Law of Thermodynamics

Statement of the Second Law

For any spontaneous process, the entropy of the universe increases:

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

- ▶ $\Delta S_{\text{univ}} > 0$: process is **spontaneous**.
- ▶ $\Delta S_{\text{univ}} < 0$: process is **nonspontaneous** (the reverse is spontaneous).
- ▶ $\Delta S_{\text{univ}} = 0$: system is at **equilibrium**.

Relating Heat and Entropy

Entropy Change for a Reversible Heat Transfer

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

where q_{rev} is heat exchanged reversibly and T is absolute temperature.

 The same heat q produces a *larger* entropy change at low T .

- ▶ Adding heat to a cold system disperses energy over *few* thermal states
 - ▶ big ΔS .
- ▶ Adding heat to a hot system disperses it over *many* thermal states
 - ▶ small ΔS .

Entropy Change of the Surroundings

Key Relation

Heat that enters the system *leaves* the surroundings, so at constant P and T :

$$\Delta S_{\text{surr}} = -\frac{q_{\text{sys}}}{T} = -\frac{\Delta H_{\text{sys}}}{T}$$

- ▶ Exothermic reactions ($\Delta H_{\text{sys}} < 0$) increase ΔS_{surr} .
- ▶ Endothermic reactions ($\Delta H_{\text{sys}} > 0$) decrease ΔS_{surr} .
- ▶ The sign of ΔS_{surr} is opposite the sign of ΔH_{sys} .

Combining the Contributions: Constant T and P

ΔS_{sys}	ΔH_{sys}	Behavior
> 0	< 0	Spontaneous at all T
< 0	> 0	Nonspontaneous at all T
> 0	> 0	Spontaneous at high T
< 0	< 0	Spontaneous at low T

Temperature can **flip** the sign of ΔS_{univ} when ΔH_{sys} and ΔS_{sys} agree in sign.

Example: Is a Phase Change Spontaneous?

Melting Ice

For $\text{H}_2\text{O}(\text{s}) \longrightarrow \text{H}_2\text{O}(\text{l})$ at 1 atm: $\Delta H_{\text{fus}} = 6.00 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S_{\text{fus}} = 22.1 \frac{\text{J}}{\text{mol K}}$.

Determine whether melting is spontaneous at (a) $T = 263 \text{ K}$ and (b) $T = 283 \text{ K}$.

$$\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$(a) \Delta S_{\text{univ}} = 22.1 - \frac{6000}{263} = -\boxed{0.7 \frac{\text{J}}{\text{mol K}}}$$

$$(b) \Delta S_{\text{univ}} = 22.1 - \frac{6000}{283} = +\boxed{0.9 \frac{\text{J}}{\text{mol K}}}$$

(a) $\Delta S_{\text{univ}} < 0$: ice does **not** melt spontaneously below 273 K.

(b) $\Delta S_{\text{univ}} > 0$: ice **does** melt spontaneously above 273 K.

► Predict ΔS_{univ} at exactly 273 K.

► What does that mean physically?

Check Your Understanding

A reaction has $\Delta S_{\text{sys}} = -50 \frac{\text{J}}{\text{K}}$ and $\Delta H_{\text{sys}} = -100 \text{ kJ}$.

At $T = 298 \text{ K}$, is the reaction spontaneous?

Come to lecture to see the answer.

Why Use A Reversible Path?

- ▶ Question
 - ▶ Why calculate entropy using a reversible path for irreversible changes?
- ▶ Evidence
 - ▶ Entropy depends only on initial and final equilibrium states.
 - ▶ Different experimental paths produce the same measured entropy change.
- ▶ Reasoning
 - ▶ $\therefore$ Microscopic accessibility depends only on the endpoints, not the transition mechanism.
 - ▶ $\therefore$ Any convenient reversible path can evaluate the same entropy change.
- ▶ Takeaway
 - ▶ State functions ignore process history.

Why Is $dS = \delta q_{\text{rev}}/T$?

- ▶ Question
 - ▶ Why does reversible heat transfer determine entropy change?
- ▶ Evidence
 - ▶ Supplying heat increases the number of accessible molecular energy states.
 - ▶ Larger microstate counts correspond to larger entropy.
- ▶ Reasoning
 - ▶ $\therefore$ Reversible heating adds energy without creating unnecessary gradients, so every increment maximally opens new accessible microstates.
 - ▶ $\therefore$ Entropy change equals the cumulative increase in accessible states quantified by $\delta q_{\text{rev}}/T$.
- ▶ Takeaway
 - ▶ Reversible heat measures microstate growth.

Why Does Temperature Appear In The Denominator?

- ▶ Question
 - ▶ Why does the same heat produce a larger entropy change at lower temperature?
- ▶ Evidence
 - ▶ Equal heat raises entropy more at 300 K than at 600 K.
 - ▶ Low-temperature systems initially occupy fewer energy levels.
- ▶ Reasoning
 - ▶ $\therefore$ Added energy opens a larger fraction of previously inaccessible states when few states were occupied initially.
 - ▶ $\therefore$ The entropy gained per unit heat decreases as temperature increases.
- ▶ Takeaway
 - ▶ Heat disperses energy most effectively when cold.

Why Does ΔS_{surr} Depend On Enthalpy?

- ▶ Question
 - ▶ Why is the surroundings entropy determined only by system enthalpy change?
- ▶ Evidence
 - ▶ At constant pressure, the system exchanges heat equal to ΔH .
 - ▶ Laboratory surroundings remain essentially at constant temperature.
- ▶ Reasoning
 - ▶ $\therefore$ A large heat reservoir changes microscopic accessibility only through the heat it receives or loses.
 - ▶ $\therefore \Delta S_{\text{surr}} = -\Delta H_{\text{sys}}/T$ for an isothermal reservoir.
- ▶ Takeaway
 - ▶ Reservoir entropy depends only on transferred heat.

Why Must Universe Entropy Increase?

- ▶ Question
 - ▶ Why is spontaneous change characterized by $\Delta S_{\text{univ}} > 0$?
- ▶ Evidence
 - ▶ Natural processes never sustain transitions toward overwhelmingly rarer macrostates.
 - ▶ Reversible processes occur only through infinitesimal equilibrium differences.
- ▶ Reasoning
 - ▶ $\therefore$ Spontaneous change increases the total accessible microstates of system plus surroundings.
 - ▶ $\therefore$ Only the reversible limit leaves total accessible microstates unchanged, giving $\Delta S_{\text{univ}} = 0$.
- ▶ Takeaway
 - ▶ Spontaneity increases universal microstate accessibility.

Outline of Part III: The Second and Third Laws of Thermodynamics

The Second Law of Thermodynamics

The Third Law and Standard Entropies

The Third Law of Thermodynamics

Statement of the Third Law

The entropy of a **perfect crystalline substance** at absolute zero is zero:

$$S(0 \text{ K}) = 0 \text{ for a perfect crystal}$$

- ▶ At 0 K, atoms have only one accessible microstate: $W = 1$, so $S = k_B \ln 1 = 0$.
- ▶ Gives entropy an **absolute** scale — unlike H and G , which only have Δ values.
- ▶ Enables tabulation of standard molar entropies S° .

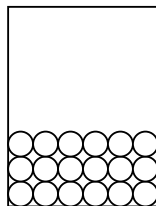
Standard Molar Entropy S°

- ▶ Different substances have different heat requirements to change their temperature and their state $\therefore$ their particles interact differently and store energy in different ways.
- ▶ As heat is absorbed, particles gain access to more positions, motions, and energy levels, increasing the number of possible microscopic arrangements.
- ▶ This increase in molecular disorder is quantified by the **standard molar entropy**, S° .

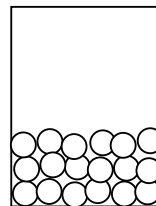
Larger molecules have higher S than smaller molecules $\therefore$ there are more parts to move.

For any given substance,

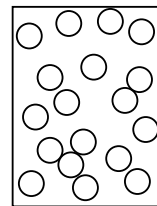
$$S^\circ(\text{solid}) < S^\circ(\text{liquid}) < S^\circ(\text{gas})$$



Solid



Liquid



Gas

Standard Molar Entropy S°

Definition

The absolute entropy of one mole of a pure substance at standard state (1 bar, specified T , usually 298 K). Units: $\text{J mol}^{-1} \text{K}^{-1}$.

- ▶ Every substance has $S^\circ > 0$ at 298 K (only 0 at 0 K).
- ▶ Gases $\gg$ liquids $>$ solids (for the same substance).
- ▶ Larger, more complex molecules have higher S° .
- ▶ Heavier atoms have higher S° than lighter atoms.

Standard Entropy Values — Some Trends

Substance	Phase	S° (J mol ⁻¹ K ⁻¹)
H ₂ O	(s)	41.0
H ₂ O	(l)	69.9
H ₂ O	(g)	188.7
CH ₄	(g)	186.4
C ₂ H ₆	(g)	229.6
Cl ₂	(g)	223.1
Br ₂	(l)	152.2
C (diamond)	(s)	2.38
C (graphite)	(s)	5.74

Notice H₂O(g) > H₂O(l) > H₂O(s); graphite > diamond (looser structure); and larger molecules have larger S° .

Standard Entropy Change of Reaction, $\Delta S_{\text{rxn}}^\circ$

Hess-Law Style Sum

For a reaction, $\Delta S_{\text{rxn}}^\circ$ is the difference between the summed S° of products and reactants:

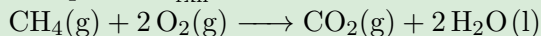
$$\Delta S_{\text{rxn}}^\circ = \sum n_p S^\circ(\text{products}) - \sum n_r S^\circ(\text{reactants})$$

- ▶ Weight each S° by its stoichiometric coefficient.
- ▶ Elemental S° is *not* zero (unlike ΔH_f° and ΔG_f°).
- ▶ Units: $\text{J mol}^{-1} \text{K}^{-1}$.

Example: Calculating $\Delta S_{\text{rxn}}^{\circ}$

Combustion of Methane

Compute $\Delta S_{\text{rxn}}^{\circ}$ at 298 K for:



Using S° (J/mol K): CH_4 186.4; O_2 205.2; CO_2 213.8; $\text{H}_2\text{O}(\text{l})$ 69.9.

$$\begin{aligned}\Delta S^{\circ} &= \left[S_{\text{CO}_2}^{\circ} + 2 S_{\text{H}_2\text{O}(\text{l})}^{\circ} \right] \\ &\quad - \left[S_{\text{CH}_4}^{\circ} + 2 S_{\text{O}_2}^{\circ} \right] \\ &= (213.8 + 2 \cdot 69.9) \\ &\quad - (186.4 + 2 \cdot 205.2) \\ &= 353.6 - 596.8\end{aligned}$$

$$\Delta S_{\text{rxn}}^{\circ} = -243.2 \frac{\text{J}}{\text{mol K}}$$

$\Delta S < 0$ — moles of gas decrease from 3 to 1, and one product is liquid.

- ▶ Would ΔS° be different if $\text{H}_2\text{O}(\text{g})$ were the product instead?
- ▶ Which sign is expected before doing arithmetic?

Why Does A Perfect Crystal Have Zero Entropy?

- ▶ Question
 - ▶ Why does a perfect crystal approach zero entropy at 0 K?
- ▶ Evidence
 - ▶ Quantum mechanics gives discrete molecular energy levels.
 - ▶ At 0 K, a perfect crystal occupies only its lowest energy state.
- ▶ Reasoning
 - ▶ $\therefore$ Thermal energy cannot populate excited states, leaving one unique microscopic arrangement.
 - ▶ $\therefore$ A perfect crystal has $W = 1$, giving zero entropy.
- ▶ Takeaway
 - ▶ Unique ground states define zero entropy.

Why Can Some Solids Retain Entropy At 0 K?

- ▶ Question
 - ▶ Why do glasses and some crystals retain entropy near 0 K?
- ▶ Evidence
 - ▶ Ice and CO crystals exhibit measurable residual entropy.
 - ▶ Different molecular orientations can possess identical minimum energies.
- ▶ Reasoning
 - ▶ $\therefore$ Multiple equally stable ground-state arrangements remain frozen when molecular motion ceases.
 - ▶ $\therefore$ More than one accessible microstate survives at 0 K.
- ▶ Takeaway
 - ▶ Ground-state degeneracy creates residual entropy.

Why Does Entropy Have An Absolute Scale?

- ▶ Question
 - ▶ Why can entropy be assigned an absolute value?
- ▶ Evidence
 - ▶ The Third Law specifies entropy at 0 K for a perfect crystal.
 - ▶ Enthalpy and Gibbs energy lack an experimentally fixed zero.
- ▶ Reasoning
 - ▶ $\therefore$ A unique microscopic reference state fixes the minimum possible entropy.
 - ▶ $\therefore$ Absolute *standard entropies* can be tabulated, unlike enthalpies.
- ▶ Takeaway
 - ▶ Microscopic reference states establish absolute entropy.

How Is Standard Entropy Measured?

- ▶ Question
 - ▶ How is absolute entropy determined experimentally?
- ▶ Evidence
 - ▶ Heat capacity is measured from near 0 K to 298 K.
 - ▶ Phase transitions absorb measurable reversible heat.
- ▶ Reasoning
 - ▶ $\therefore$ Each reversible heating step reveals additional accessible microstates accumulated from the unique ground state.
 - ▶ $\therefore$ Summing all reversible entropy increases yields the absolute standard entropy.
- ▶ Takeaway
 - ▶ Absolute entropy accumulates from reversible heating.

Why Does Graphite Have Higher Entropy Than Diamond?

- ▶ Question
 - ▶ Why is graphite's standard entropy larger than diamond's?
- ▶ Evidence
 - ▶ Both are crystalline forms of pure carbon.
 - ▶ Graphite has a larger measured standard entropy than diamond.
- ▶ Reasoning
 - ▶ $\therefore$ Graphite's layered bonding permits more low-energy lattice vibrations than diamond's rigid network.
 - ▶ $\therefore$ More vibrational microstates are populated at ordinary temperatures.
- ▶ Takeaway
 - ▶ Vibrational accessibility controls crystalline entropy.

Check Your Understanding

Rank in order of *increasing* standard molar entropy:
 $\text{H}_2\text{O}(\text{s})$, $\text{H}_2\text{O}(\text{l})$, $\text{H}_2\text{O}(\text{g})$, $\text{NaCl}(\text{s})$, $\text{HCl}(\text{g})$

Come to lecture to see the answer.

Part IV

Free Energy

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

Gibbs Free Energy G – the Criterion for Spontaneity

Definition (J. W. Gibbs, 1876)

$$G = H - TS$$

At constant T and P ,

$$\Delta G = \Delta H - T\Delta S$$

- ▶ G combines H and S into a single spontaneity criterion.
- ▶ ΔG refers only to the **system**, unlike ΔS_{univ} .
- ▶ Units of ΔG : kJ/mol (matching ΔH).

Deriving ΔG from ΔS_{univ}

From the Second Law

Start with $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$; at constant T and P :

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T}$$

$$-T\Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}}$$

$$\boxed{\Delta G_{\text{sys}} = -T\Delta S_{\text{univ}}}$$

Because $T > 0$, the sign of ΔG_{sys} is *opposite* to ΔS_{univ} .

ΔG and Spontaneity

Spontaneity Criterion (constant T, P)

$\Delta G < 0$ Spontaneous (exergonic)

$\Delta G > 0$ Nonspontaneous (endergonic)

$\Delta G = 0$ At equilibrium

- ▶ A single number (ΔG) indicates the direction of a process.
- ▶ We can now stay entirely within the system — no need to compute ΔS_{surr} .
- ▶ G is the “free” energy available to do non- PV work.

Examples of non- PV work include: producing electrical work in a battery, driving an electrical current, and certain forms of mechanical or chemical work.

Standard Free Energy of Reaction $\Delta G_{\text{rxn}}^{\circ}$

Two Equivalent Routes

$$\Delta G_{\text{rxn}}^{\circ} = \Delta H_{\text{rxn}}^{\circ} - T \Delta S_{\text{rxn}}^{\circ}$$

$$\Delta G_{\text{rxn}}^{\circ} = \sum n_p \Delta G_f^{\circ}(\text{products}) - \sum n_r \Delta G_f^{\circ}(\text{reactants})$$

- ▶ ΔG_f° = standard free energy of formation (from elements).
- ▶ $\Delta G_f^{\circ} = 0$ for a pure element in its standard state.
- ▶ The two routes give the same answer (Hess's Law generalized).

Example: Two Ways to Compute $\Delta G_{\text{rxn}}^{\circ}$

Combustion of Methane

For $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ at 298 K:

$$\Delta H_{\text{rxn}}^{\circ} = -890.4 \frac{\text{kJ}}{\text{mol}}, \Delta S_{\text{rxn}}^{\circ} = -243.2 \frac{\text{J}}{\text{mol K}}.$$

Method 1: use ΔH° and ΔS° .

$$\begin{aligned}\Delta G^{\circ} &= \Delta H^{\circ} - T\Delta S^{\circ} \\ &= -890.4 - (298)(-0.2432) \\ &= -890.4 + 72.5\end{aligned}$$

$$\Delta G^{\circ} = -817.9 \frac{\text{kJ}}{\text{mol}}$$

$\Delta G^{\circ} < 0 \Rightarrow$ combustion of methane is **spontaneous** at 298 K.

The large exothermic ΔH° overwhelms the negative ΔS° .

- ▶ How would using tabulated ΔG_f° values compare?
- ▶ Why is the entropy term written in kJ, not J?

Example: Using ΔG_f° Values

Same Reaction, Different Route

For $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ use

ΔG_f° (kJ/mol): $\text{CH}_4 = -50.5$; $\text{CO}_2 = -394.4$; $\text{H}_2\text{O}(\text{l}) = -237.1$; $\text{O}_2 = 0$.

$$\begin{aligned}\Delta G^\circ &= [\Delta G_f^\circ(\text{CO}_2) + 2 \Delta G_f^\circ(\text{H}_2\text{O}(\text{l}))] \\ &\quad - [\Delta G_f^\circ(\text{CH}_4) + 2 \Delta G_f^\circ(\text{O}_2)] \\ &= [-394.4 + 2(-237.1)] - [-50.5 + 0] \\ &= -868.6 + 50.5\end{aligned}$$

$$\Delta G^\circ = -818.1 \frac{\text{kJ}}{\text{mol}}$$

Agrees with the $\Delta H^\circ - T\Delta S^\circ$ route within rounding.

Choose whichever data are more readily available.

- ▶ Why is $\Delta G_f^\circ(\text{O}_2) = 0$?
- ▶ Is the reaction spontaneous?

Why Invent Gibbs Free Energy?

- ▶ Question
 - ▶ Why replace the universe entropy criterion with a new system property?
- ▶ Evidence
 - ▶ Spontaneity requires evaluating $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$.
 - ▶ Surroundings are rarely measured directly in laboratory experiments.
- ▶ Reasoning
 - ▶ $\therefore$ The surroundings behave predictably as an isothermal, constant-pressure reservoir.
 - ▶ $\therefore$ Their contribution can be incorporated into a single *system* state function.
- ▶ Takeaway
 - ▶ Gibbs energy embeds surroundings into system analysis.

Why Is Gibbs Energy Defined At Constant T And P ?

- ▶ Question
 - ▶ Why does Gibbs energy predict spontaneity only at constant temperature and pressure?
- ▶ Evidence
 - ▶ The surroundings entropy relation uses constant reservoir temperature.
 - ▶ Enthalpy equals exchanged heat only at constant pressure.
- ▶ Reasoning
 - ▶ $\therefore$ Both substitutions fail if temperature or pressure changes during the process.
 - ▶ $\therefore$ Only constant T, P allow Gibbs energy to replace universe entropy.
- ▶ Takeaway
 - ▶ Thermodynamic potentials require specific constraints.

Why Does $\Delta G = -T\Delta S_{\text{univ}}$?

- ▶ Question
 - ▶ How does Gibbs energy become equivalent to the Second Law?
- ▶ Evidence
 - ▶ $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$.
 - ▶ $\Delta S_{\text{surr}} = -\Delta H_{\text{sys}}/T$ for an ideal reservoir.
- ▶ Reasoning
 - ▶ $\therefore$ Eliminating the surroundings leaves $\Delta H_{\text{sys}} - T\Delta S_{\text{sys}} = -T\Delta S_{\text{univ}}$.
 - ▶ $\therefore$ Defining $G = H - TS$ makes one system quantity report universal spontaneity.
- ▶ Takeaway
 - ▶ Gibbs energy reframes the Second Law.

Why Does A Negative ΔG Mean Spontaneous?

- ▶ Question
 - ▶ Why does Gibbs energy decrease when universe entropy increases?
- ▶ Evidence
 - ▶ Absolute temperature is always greater than zero.
 - ▶ Spontaneous processes satisfy $\Delta S_{\text{univ}} > 0$.
- ▶ Reasoning
 - ▶ $\therefore$ Multiplying by the positive factor $-T$ reverses the inequality without changing its magnitude.
 - ▶ $\therefore \Delta G < 0$ exactly matches the Second Law criterion for spontaneity.
- ▶ Takeaway
 - ▶ Gibbs energy inherits the Second Law's direction.

Why Is Gibbs Energy Called Free Energy?

- ▶ Question
 - ▶ Why does Gibbs energy represent energy available for useful work?
- ▶ Evidence
 - ▶ Some released energy must enter the surroundings during spontaneous change.
 - ▶ Entropy production accompanies every irreversible process.
- ▶ Reasoning
 - ▶ $\therefore$ Increasing surroundings entropy requires unavoidable thermal energy dispersal that cannot perform directed work.
 - ▶ $\therefore$ The remaining decrease in Gibbs energy equals the maximum non-expansion work available.
- ▶ Takeaway
 - ▶ Entropy production limits useful energy.

Why Do Different Constraints Need Different Potentials?

- ▶ Question
 - ▶ Why use Helmholtz energy at constant T, V but Gibbs energy at constant T, P ?
- ▶ Evidence
 - ▶ Constant-volume systems perform no expansion work.
 - ▶ Constant-pressure systems exchange expansion work with the surroundings.
- ▶ Reasoning
 - ▶ $\therefore$ Each constraint fixes different unavoidable energy exchanges with the surroundings.
 - ▶ $\therefore$ Thermodynamic potentials subtract unavoidable contributions, leaving only energy available to drive change.
- ▶ Takeaway
 - ▶ Potentials reflect experimental constraints.

Check Your Understanding

A reaction has $\Delta H^\circ = +40.0 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +50.0 \frac{\text{J}}{\text{mol K}}$.

Compute ΔG° at 298 K. Is the reaction spontaneous at this temperature?

Come to lecture to see the answer.

Outline of Part IV: Free Energy

Gibbs Free Energy

Temperature Dependence of Spontaneity

Free Energy and Equilibrium

Chapter Summary

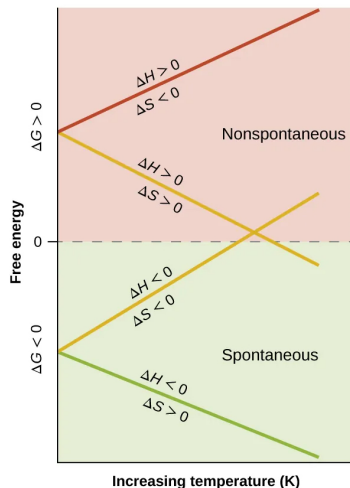
Four Combinations of ΔH and ΔS

Summary of the Four Scenarios for Enthalpy and Entropy Changes

	$\Delta H > 0$ (endothermic)	$\Delta H < 0$ (exothermic)
$\Delta S > 0$ (increase in entropy)	$\Delta G < 0$ at high temperature $\Delta G > 0$ at low temperature Process is spontaneous at high temperature	$\Delta G < 0$ at any temperature Process is spontaneous at any temperature
$\Delta S < 0$ (decrease in entropy)	$\Delta G > 0$ at any temperature Process is nonspontaneous at any temperature	$\Delta G < 0$ at low temperature $\Delta G > 0$ at high temperature Process is spontaneous at low temperature

The sign of $\Delta G = \Delta H - T\Delta S$ depends jointly on the signs of ΔH , ΔS , and on temperature.

ΔG vs. Temperature for Each Case



- **Green:** $\Delta H < 0$, $\Delta S > 0$ — always spontaneous.
- **Red:** $\Delta H > 0$, $\Delta S < 0$ — never spontaneous.
- **Yellow (up):** $\Delta H > 0$, $\Delta S > 0$ — spontaneous only above a crossover T .
- **Yellow (down):** $\Delta H < 0$, $\Delta S < 0$ — spontaneous only below a crossover T .

Crossover Temperature

At Equilibrium, $\Delta G = 0$

Setting $\Delta G = \Delta H - T\Delta S = 0$ and solving:

$$T_{\text{eq}} = \frac{\Delta H}{\Delta S}$$

- ▶ T_{eq} is the temperature at which the reaction switches direction.
- ▶ Above T_{eq} the sign of ΔG flips.
- ▶ Use ΔH and ΔS in *matching* units!

Example: Boiling Point from Thermodynamics

Boiling Point of Ethanol

For $\text{C}_2\text{H}_5\text{OH}(\text{l}) \longrightarrow \text{C}_2\text{H}_5\text{OH}(\text{g})$: $\Delta H^\circ = +42.4 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +119 \frac{\text{J}}{\text{mol K}}$.

Estimate the normal boiling point of ethanol.

$\Delta G^\circ = 0$ at boiling

$$\begin{aligned} T_{\text{bp}} &= \frac{\Delta H^\circ}{\Delta S^\circ} \\ &= \frac{42\,400 \frac{\text{J}}{\text{mol}}}{119 \frac{\text{J}}{\text{mol K}}} \end{aligned}$$

$$T_{\text{bp}} \approx 356 \text{ K} \approx 83^\circ \text{C}$$

The measured boiling point of ethanol is 78°C — a very good estimate from ΔH° and ΔS° alone.

- Why is the estimate a bit high?
- Below T_{bp} , is vaporization spontaneous?

Why Does ΔG Vary Linearly with Temperature?

- ▶ Question
 - ▶ Why does changing temperature shift ΔG almost linearly?
- ▶ Evidence
 - ▶ Measured ΔH and ΔS change little over modest temperature ranges.
 - ▶ Experimental ΔG - T plots are nearly straight lines.
- ▶ Reasoning
 - ▶ $\therefore$ Similar molecular motions and bonding keep *enthalpy* and *entropy* nearly constant as temperature changes modestly.
 - ▶ $\therefore \Delta G = \Delta H - T\Delta S$ changes linearly with slope $-\Delta S$ until heat capacities become important.
- ▶ Takeaway
 - ▶ Constant molecular responses produce nearly linear $\Delta G(T)$.

The Physical Meaning of the Crossover Temperature

- ▶ Question
 - ▶ Why does one temperature separate spontaneous and nonspontaneous behavior?
- ▶ Evidence
 - ▶ At the crossover, experiments find $\Delta G = 0$.
 - ▶ Solving gives $T_{\text{eq}} = \Delta H / \Delta S$ when $\Delta S \neq 0$.
- ▶ Reasoning
 - ▶ $\therefore$ Raising temperature strengthens the entropy contribution by multiplying every additional accessible microstate by T .
 - ▶ $\therefore$ At T_{eq} , $T\Delta S$ exactly cancels ΔH , so neither energetic tendency dominates.
- ▶ Takeaway
 - ▶ Temperature tunes competition between energy and entropy.

Why Is T_{eq} a Phase Transition Temperature?

- ▶ Question
 - ▶ Why does T_{eq} become the melting or boiling temperature?
- ▶ Evidence
 - ▶ At melting and boiling points, both phases coexist without net change.
 - ▶ Molecules continuously transfer between phases in both directions.
- ▶ Reasoning
 - ▶ $\therefore$ Equal molecular tendencies make forward and reverse transfers equally probable.
 - ▶ $\therefore$ The phases remain in *equilibrium*, so $\Delta G = 0$ identifies the transition temperature.
- ▶ Takeaway
 - ▶ Equilibrium reflects balanced molecular exchange.

Why Must ΔH and ΔS Have the Same Sign?

- ▶ Question
 - ▶ Why do only matching signs produce a crossover temperature?
- ▶ Evidence
 - ▶ A positive temperature requires $T_{\text{eq}} = \Delta H / \Delta S > 0$.
 - ▶ Opposite signs make the ratio negative.
- ▶ Reasoning
 - ▶ $\therefore$ Enthalpy and entropy then favor the same direction, so temperature cannot reverse their combined effect.
 - ▶ $\therefore$ Spontaneity remains unchanged across all physically accessible temperatures.
- ▶ Takeaway
 - ▶ Temperature only shifts competing driving forces.

Why Does Ice Melt Above the Transition Temperature?

- ▶ Question
 - ▶ Why does ice melt spontaneously only above T_{eq} ?
- ▶ Evidence
 - ▶ Melting absorbs heat and greatly increases accessible molecular arrangements.
 - ▶ Above the melting point, liquid water forms without continued external forcing.
- ▶ Reasoning
 - ▶ $\therefore$ Water molecules gain many more accessible microstates than the surroundings lose when supplying heat.
 - ▶ $\therefore$ Above T_{eq} , the entropy benefit outweighs the energetic cost, making melting spontaneous.
- ▶ Takeaway
 - ▶ Entropy dominates once temperature amplifies its influence.

Check Your Understanding

The reaction $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ has $\Delta H^\circ = +178 \frac{\text{kJ}}{\text{mol}}$ and $\Delta S^\circ = +161 \frac{\text{J}}{\text{mol K}}$.

Above what temperature does CaCO_3 spontaneously decompose?

Come to lecture to see the answer.

Outline of Part IV: Free Energy

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Free Energy Away from Standard Conditions

Non-Standard Free Energy Change

For any composition, ΔG depends on the reaction quotient Q :

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where $R = 8.314 \frac{\text{J}}{\text{mol K}}$.

- ▶ ΔG° is the value at standard state ($Q = 1$).
- ▶ Q is the same reaction quotient used with K in Ch. 13.
- ▶ ΔG (not ΔG°) determines actual spontaneity.

ΔG° and K at Equilibrium

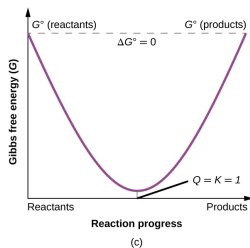
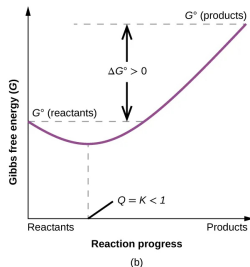
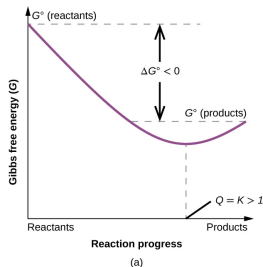
At Equilibrium, $\Delta G = 0$ and $Q = K$

$$0 = \Delta G^\circ + RT \ln K$$

$$\boxed{\Delta G^\circ = -RT \ln K}$$

- ▶ $\Delta G^\circ < 0 \Rightarrow K > 1$ (products favored).
- ▶ $\Delta G^\circ > 0 \Rightarrow K < 1$ (reactants favored).
- ▶ $\Delta G^\circ = 0 \Rightarrow K = 1$.

Reaction Progress and Free Energy



A reaction proceeds “downhill” in G to the composition where $\Delta G = 0$. The horizontal position of that minimum determines K .

(a) $\Delta G^\circ < 0 \Rightarrow K > 1$

(b) $\Delta G^\circ > 0 \Rightarrow K < 1$

(c) $\Delta G^\circ = 0 \Rightarrow K = 1$

Interpreting the Reaction Coordinate

The equilibrium composition sits at the *minimum* of G vs. progress:

- ▶ A large negative ΔG° pushes the minimum toward products.
- ▶ A large positive ΔG° pushes it toward reactants.
- ▶ Reactions do not always run “to completion” — they run to the minimum of G .

Even for very negative ΔG° , a tiny amount of reactant remains at equilibrium.

Example: ΔG° from K

Free Energy $\leftrightarrow$ Equilibrium Constant

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ at 298 K, $K_p = 0.144$.

Compute ΔG° .

$$\begin{aligned}\Delta G^\circ &= -RT \ln K \\ &= -(8.314)(298) \ln(0.144) \\ &= -(2478)(-1.938)\end{aligned}$$

$$\Delta G^\circ \approx +4.80 \frac{\text{kJ}}{\text{mol}}$$

$$K < 1 \Rightarrow \Delta G^\circ > 0.$$

The equilibrium mixture at 298 K favors N_2O_4 .

- What sign of ΔG° would $K = 1$ give?
- How could raising T shift the equilibrium here?

Composition Changes the Driving Force

- ▶ Question
 - ▶ Why does ΔG depend on composition as a reaction proceeds?
- ▶ Evidence
 - ▶ Measured reaction driving force changes although temperature remains constant.
 - ▶ Species chemical potentials follow $\mu_i = \mu_i^\circ + RT \ln a_i$ for ideal behavior.
- ▶ Reasoning
 - ▶ $\therefore$ Mixing changes each species' molecular surroundings, altering escaping tendency through its activity.
 - ▶ $\therefore$ Summing reactant and product chemical potentials gives
$$\Delta G = \Delta G^\circ + RT \ln Q.$$
- ▶ Takeaway
 - ▶ Composition continuously reshapes the reaction driving force.

Why Does the Logarithm Appear?

- ▶ Question
 - ▶ Why is composition dependence logarithmic rather than linear?
- ▶ Evidence
 - ▶ Mixing identical gases increases entropy without changing intermolecular energy.
 - ▶ Statistical entropy satisfies $S = k \ln W$.
- ▶ Reasoning
 - ▶ $\therefore$ Independent molecular arrangements multiply, while logarithms convert multiplicative changes into additive entropy contributions.
 - ▶ $\therefore$ Composition enters chemical potential and ΔG through logarithms of activity.
- ▶ Takeaway
 - ▶ Logarithms arise from counting molecular arrangements.

Why Is Equilibrium a Minimum in Gibbs Energy?

- ▶ Question
 - ▶ Why does equilibrium occur at minimum Gibbs energy instead of $\Delta G^\circ = 0$?
- ▶ Evidence
 - ▶ Measured G decreases until reaction progress stops.
 - ▶ ΔG° remains fixed at constant temperature.
- ▶ Reasoning
 - ▶ $\therefore$ Composition changes during reaction, continually changing ΔG while ΔG° stays unchanged.
 - ▶ $\therefore$ Reaction stops only where further composition changes cannot lower G .
- ▶ Takeaway
 - ▶ Equilibrium minimizes Gibbs energy, not standard Gibbs energy.

How Does the Equilibrium Constant Emerge?

- ▶ Question
 - ▶ How does the minimum in Gibbs energy determine K ?
- ▶ Evidence
 - ▶ At equilibrium, $\Delta G = 0$ and $Q = K$.
 - ▶ Activities have one unique equilibrium ratio at fixed temperature.
- ▶ Reasoning
 - ▶ $\therefore$ Equilibrium fixes the composition where energetic and entropic tendencies exactly balance.
 - ▶ $\therefore$ Substituting equilibrium conditions gives $\Delta G^\circ = -RT \ln K$.
- ▶ Takeaway
 - ▶ Standard driving force determines equilibrium composition.

Why Is K So Sensitive to ΔG° ?

- ▶ Question
 - ▶ Why do small changes in ΔG° cause huge changes in K ?
- ▶ Evidence
 - ▶ Changing ΔG° by about RT changes K by a factor of e .
 - ▶ Equilibrium constants span many orders of magnitude.
- ▶ Reasoning
 - ▶ $\therefore$ Multiplicative changes in accessible equilibrium compositions become additive in Gibbs energy through logarithms.
 - ▶ $\therefore$ Small energy differences produce exponentially different equilibrium compositions.
- ▶ Takeaway
 - ▶ Energy differences amplify into large composition changes.

Why Does Equilibrium Never Mean Complete Conversion?

- ▶ Question
 - ▶ Why does favorable chemistry still leave some reactant?
- ▶ Evidence
 - ▶ Even strongly favorable reactions contain measurable reactant at equilibrium.
 - ▶ The G -reaction coordinate curve has a minimum, not a cliff.
- ▶ Reasoning
 - ▶ $\therefore$ Converting additional reactant eventually increases Gibbs energy once the minimum is passed.
 - ▶ $\therefore$ Forward and reverse reactions balance before complete conversion occurs.
- ▶ Takeaway
 - ▶ Minimum energy fixes a mixed equilibrium state.

Why Does Temperature Change the Equilibrium Constant?

- ▶ Question
 - ▶ Why must changing temperature shift the equilibrium constant?
- ▶ Evidence
 - ▶ Heating changes measured equilibrium compositions.
 - ▶ $\ln K$ varies systematically with temperature.
- ▶ Reasoning
 - ▶ $\therefore$ Temperature changes the balance between energetic stabilization and entropic stabilization established earlier.
 - ▶ $\therefore$ The equilibrium composition shifts, giving a new constant K at each temperature.
- ▶ Takeaway
 - ▶ Temperature moves equilibrium by changing thermodynamic balance.

Check Your Understanding

For a reaction at 298 K, $\Delta G^\circ = -34.3 \frac{\text{kJ}}{\text{mol}}$.

Estimate the value of the equilibrium constant K .

Come to lecture to see the answer.

Outline of Part IV: Free Energy

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Chapter Summary

Chapter 16 Key Equations

Boltzmann's Equation

$$S = k_B \ln W$$

Reversible Heat and Entropy

$$\Delta S = q_{\text{rev}}/T$$

Surroundings

$$\Delta S_{\text{surr}} = -\Delta H_{\text{sys}}/T$$

Second Law

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

Free Energy of Reaction

$$\Delta G^\circ = \sum n_p \Delta G_f^\circ - \sum n_r \Delta G_f^\circ$$

Non-Standard ΔG

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Free Energy & K

$$\Delta G^\circ = -RT \ln K$$

Takeaways

- ▶ Spontaneous processes disperse **matter** and **energy**.
- ▶ Entropy S counts microstates: $S = k_B \ln W$.
- ▶ Second Law: $\Delta S_{\text{univ}} > 0$ for any spontaneous change.
- ▶ Third Law: $S = 0$ at 0 K for a perfect crystal.
- ▶ Free energy $\Delta G = \Delta H - T\Delta S$ determines whether a process is spontaneous.
- ▶ At equilibrium, $\Delta G = 0$ and $\Delta G^\circ = -RT \ln K$.

Thermodynamics tells us *whether* a reaction is thermodynamically favorable (spontaneous) by considering both energy and entropy through the Gibbs free energy, ΔG . It indicates *if a reaction can occur*.

Thermochemistry (ΔH) tells us the heat change of a reaction — whether heat is released (exothermic, $\Delta H < 0$) or absorbed (endothermic, $\Delta H > 0$), and by how much.

Kinetics tells us *how fast a reaction occurs* by examining the reaction rate and the activation energy. A reaction may be thermodynamically favorable yet still proceed very slowly if it has a large activation energy barrier.